ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/326579730

Binomial Models And Option Pricing valuation in R

Article · July 2018

CITATIONS
0

READS
6,607

1 author:

Event Makamo
University of South Africa

10 PUBLICATIONS **0** CITATIONS

SEE PROFILE

Some of the authors of this publication are also working on these related projects:

Project Financial Risk and Volatility forecasting View project

All content following this page was uploaded by Event Makamo on 25 July 2018.

The user has requested enhancement of the downloaded file.

Binomial Models And Option Pricing valuation in R

Event S. Makamo

sanelemakamo26@gmail.com

July 25, 2018

Contents

1 Introduction **2**

2 Binomial experiments **2**

3 Binomial Trees **3**

4 Option Pricing **5**

5 Put-Call Parity Model **8**

6 Valuing Options through a replicating portfolio **9**

6.1 One period case 9

6.2 Two-period case 12

7 Arbitrage Strategy **15**

8 Conclusion **16**

9 references **17**

1 Introduction

In this short paper we are going to explore the use of binomial trees in option pricing using R. R is an open source statistical softaware program that can be downloaded for free at *www.r-project.org*.
We first begin with the formal definition of binomial experiments and extend that to binomial trees. We then move on to model option pricing using binomial trees in R and the valuation of options through a replicating portfolio. We end with the arbitrage strategy.

2 Binomial experiments

Coin tossing is oftenly used to model the movement of share prices, in which the successive tosses represent a binomial trial with two outcomes, 'success' or 'failure'. If a coin is tossed twice say $X = \{H,T\}^2$. The probability of getting a head(H) is denoted by p and the probability of getting a tail is denoted by (1-p).
The number of successive tosses is denoted by n which follows a discrete process. The binomial probability distribution of X is given by,

$$f_X(x) = \sum_{k=1}^{n} p^k (1-p)^{n-k}$$

In the case where n=2 , the probability space is $(\alpha,P) = \{w_1,w_2,w_3,w_4\}$ were $w_1 = \{HH\}$,$w_2 = \{HT\}$,$w_3 = \{TH\}$ and $w_4 = \{TT\}$
The probability of success is p=0.5. This is known as a binomial experiment. The following sections extends the binomial trial above to the pricing of options using the Binomial trees.

3 Binomial Trees

Binomial tree is a technique used to model the movement of stock prices. It is a diagram that depicts the up-move and the down-move of the share price. Each step is accompanied by a probability of an upward movement and a downward movement.

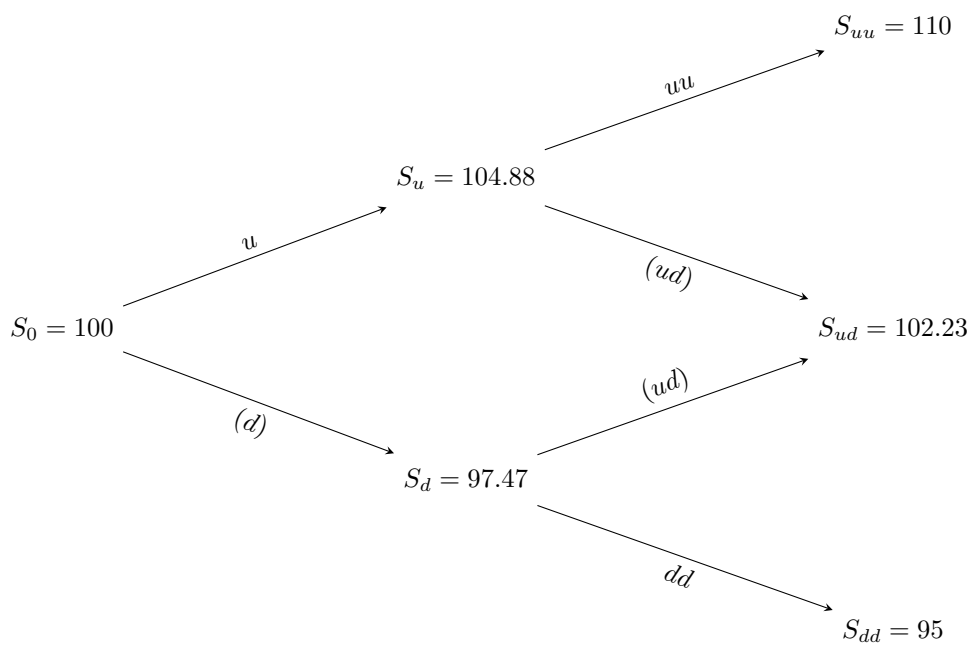
For illustration, lets consider the successive coin tosses explained above as a share price. Lets assume the starting price of the share is $S_(0) = \$100$, the strike price is $K = \$100$, the risk-free rate is $r = 2.5\%$, $t = 0,1,2$. The upward move and downward move of the share is $u = 1.0488$ and $d = 0.9747$, respectively.

To calculate the outcome of the share price at each step, we write,
At t=0, we have $S_0 = \$100$

At t = 1, $S_u = uS_0 = (1.0488)(100) = \$104.88$, similarly $S_d = (0.9747)(100) = \$97.47$

At t=2, $S_{uu} = uuS_0 = \$110$, $S_{ud} = udS_0 = \$102.23$ and $S_{dd} = ddS_0 = \$95$

The 2-step binomial tree diagram is depicted as,



The R code for the above binomial tree is given by

```
stock_tree = function(S, delta_t,u,d, N) {
  tree = matrix(0, nrow=N+1, ncol=N+1)

  sigma=log(u)/sqrt(delta_t)
  sigma=log(d)/sqrt(delta_t)

  for (i in 1:(N+1)) {
    for (j in 1:i) {
      tree[i,j] = S * u^(j-1) * d^((i-1)-(j-1))
    }
  }
  return(tree)
}
stock_tree(100,1,1.0488,0.9747,2)
```

The binomial tree diagram in R is given by

```
> stock_tree(100,1,1.0488,0.9747,2)
          [,1]     [,2]     [,3]
[1,] 100.00000   0.0000   0.0000
[2,]  97.47000 104.8800   0.0000
[3,]  95.00401 102.2265 109.9981
```

The results correspond with those we got above in our binomial tree diagram.

4 Option Pricing

Consider a situation of an investor who buys a European call option with a strike price of \$ 100 to purchase 100 shares of a particular stock. If the current share price is \$100 with a specified expiration date, because the option is European, the investor can exercise only on the expiration date. If the stock price on this date is less than \$ 100, the investor will decide not to exercise the option. However, if the stock price is above \$ 100 on the expiration date, the investor may exercise the option. Suppose, for example, that the share price is \$ 100.

The risk-neutral probability of an up-move is $q = \frac{(1+r)\triangle t - d}{u-d}$ and if the short rate r is continuously compounded , $q = \frac{e^{r\triangle t} - d}{u-d}$ and the risk neutral probability of a down-move is $(1-q)$ were $\triangle t = \frac{T}{N}$.

The price of the European call option C_0 and the price of a European put option P_0 is given by the following formula also known as the CRR model.

$$C_0 = (e^{-r\triangle t})^N \sum_{k=0}^{N} \frac{N!}{k!(N-k)!}(S_0 u^k d^{N-k} - K)^+ q^k (1-q)^{N-K}$$

$$P_0 = (e^{-r\triangle t})^N \sum_{k=0}^{N} \frac{N!}{k!(N-k)!}(K - S_0 u^k d^{N-k})^+ q^k (1-q)^{N-K}$$

To build a European option tree, we first calculate the risk neutral probability,

$$q = \frac{(1+0.025)\frac{2}{2} - 0.98}{1.0488 - 0.9747} = 0.6788$$

The Rcode for calculating the risk neutral probability q is

```
q_prob = function(r, delta_t,u,d) {
  X=((1+r)*delta_t)
  return(((X*delta_t) - d)/(u-d))
}
q_prob(0.025,1,1.0488,0.9747)
```

In R we get the following results which corresponds with our calculations of the risk-neutral probability q.

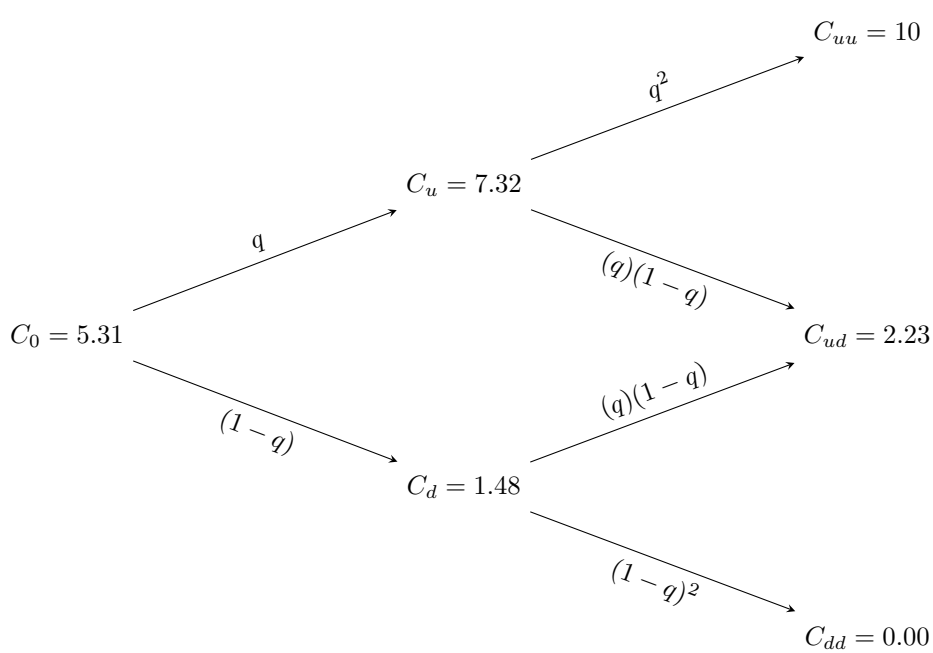
```
> q_prob(0.025,1,1.0488,0.9747)
[1] 0.6788124
```

The equivalent R code for the above CRR model which accounts for both the European call and European put options is as follows,

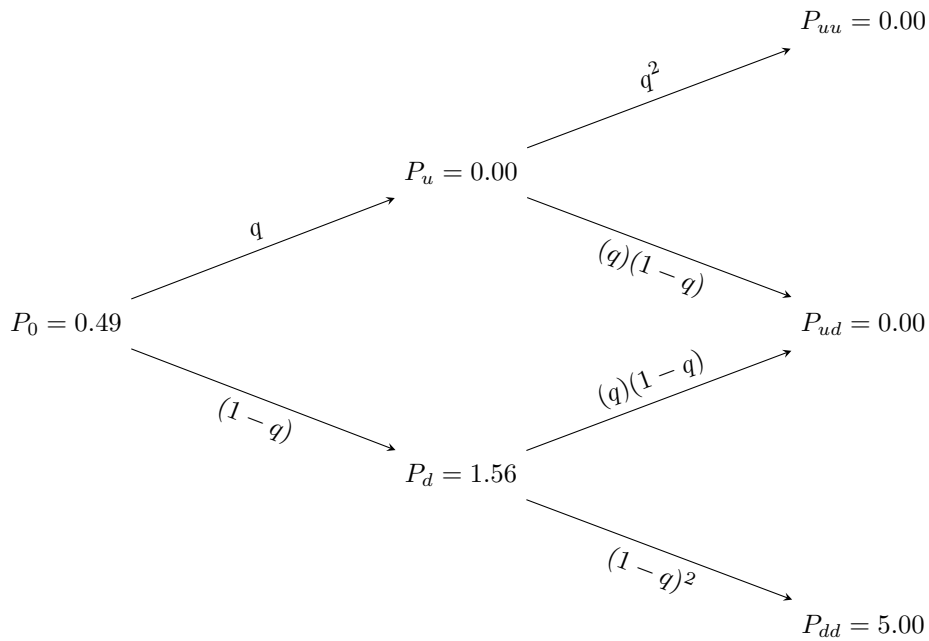
```
CRR_model_call_put = function(tree,u,d, delta_t, r, K, type) {
  q = q_prob(r, delta_t,u,d)

  option_tree = matrix(0, nrow=nrow(tree), ncol=ncol(tree))
  if(type == 'put') {
    option_tree[nrow(option_tree),] = pmax(K - tree[nrow(tree),], 0)
  } else {
    option_tree[nrow(option_tree),] = pmax(tree[nrow(tree),] - K, 0)
  }

  for (i in (nrow(tree)-1):1) {
    for(j in 1:i) {
      option_tree[i, j] = ((1-q)*option_tree[i+1,j] + q*option_tree[i+1,j+1])/exp(r*delta_t)
    }
  }
  return(option_tree)
}
tree=stock_tree(100,1,1.0488,0.9747,2)
CRR_model_call_put(tree, 1.0488,0.9747,1,0.025,100,'put')
```

We get the following 2-step European call option binomial tree

```
> CRR_model_call_put(tree, 1.0488,0.9747,1,0.025,100,'call')
         [,1]     [,2]     [,3]
[1,] 5.305857 0.000000 0.000000
[2,] 1.474084 7.316775 0.000000
[3,] 0.000000 2.226536 9.998144
```



The 2-Step European put option tree is



The corresponding CRR model results in R is as follows

```
> CRR_model_call_put(tree, 1.0488,0.9747,1,0.025,100,'put')
          [,1] [,2] [,3]
[1,] 0.4902577    0    0
[2,] 1.5650313    0    0
[3,] 4.9959910    0    0
```

5 Put-Call Parity Model

The put-call parity model is a model that indicates the equilibrium relationship between prices of a call and put options on the same stock. In this model, the equilibrium price of a European put is determined by the current value of a portfolio consisting of a long position in the call, a long position in the riskless bond with a face value equal to the exercise price, and the short position in the stock:

$$P_0 = C_0 + PV(X) - S_0$$

were $PV(X) = \frac{S_0}{(1+r)^N}$

We can use this model to determine the equilibrium price of the put that we obatianed using the binomial put model by substituting the equilibrium call price as determined by the binomial call model.

$$P_0 = 5.31 + \frac{100}{(1.04)^2} - 100 = \$0.49$$

The put-call parity model in R is given by,

```
put_call_parity = function(ca, pa, S, r, N, type){
  if(type == 'put'){
    parity = ca + (S/(1+r)^N) - S
  } else{
    parity = pa - (S/(1+r)^N) + S
  }

  return(parity)
}
put_call_parity(5.31,0.49, 100, 0.025, 2, 'put')
```

The equilibrium put/call price at t=0 in R is given by:

```
> put_call_parity(5.31,0.49, 100, 0.025, 2, 'put')
[1] 0.4914396
> put_call_parity(5.31,0.49, 100, 0.025, 2, 'call')
[1] 5.30856
```

We see that the R code gives us the same results as the above put-call parity model.

6 Valuing Options through a replicating portfolio

In the binomial pricing model , the equilibrium price of an option is underpinned on the law of a one price. In the case of a call, this one price is evaluated by equating the price of the call to the value of a replicating portfolio. A replicating portfolio is a portfolio constructed such that its possible cash flows are equal to the call's possible payouts.

A replicating portfolio whose outflows at the end of the period exactly match the call's can be formed by buying H_0 shares of stock at a price of S_0 partially financed by borrowing B_0 dollars at a risk-free rate, that is, selling the risk-free security short. The current value of this portfolio , V_0 , is

$$V_0 = H_0 S_0 - B_0$$

where the negative sign means borrowing and, contigent on the future state, at the end of the period , the portfolio will have one of the following states:

$$V_u = H_0 u S_0 - r B_0$$

$$V_d = H_0 d S_0 - r B_0$$

Given these two possible values, the replicating portfolio can be formed by finding the unknowns, H_0,B_0, which make two possible portfolio values, V_u and V_d, equal to two possible call values, C_u and C_d in their respective states.

$$C_u = H_0 u S_0 - r B_0$$
$$C_d = H_0 d S_0 - r B_0$$

Solving these simulteneously, we have

6.1 One period case

$$H_0 = \frac{C_u - C_d}{uS_0 - dS_0}$$

$$B_0 = \frac{C_u(dS_0 - C_d(uS_0)}{r(uS_0 - dS_0)}$$

For the binomial European call option with $C_u = \$7.32$,$C_d = 1.48u = 1.0488$,$d = 0.9747$,$r = 0.025$,$S_0 = \$100$ we have

$$H_0 = \frac{7.32 - 1.48}{104.88 - 97.47} = 0.7881$$

$$B_0 = \frac{(7.32)(97.47) - (1.48)(104.88)}{(1.025)[104.88 - 97.47]} = \$73.501$$

H_0 is often referred as the *hedge ratio or delta value*. Thus, to replicate the two possible call payouts, an investor would need to buy 0.7881 shares of stock at \$100 per share, partially financed by borrowing \$73.501. At the end of the period, we have

$$V_u = (0.7881)(1.0488)(100) - (1.025)(73.501) = \$7.32$$

$$V_d = (0.7881)(0.9747)(100) - (1.025)(73.501) = \$1.48$$

Which replicates the two possible call payoffs. Lastly, by the law of the single price, we can determine the equilibrium price of the call, C_0, by setting the current call value equal to the current value of the replicating portfolio.

$$C_0 = H_0 S_0 - B_0 = (0.7881)(100) - 73.501 = \$5.31$$

The corresponding in R for H_0 for both the call and the put is given below :

```
call_replicate_porfolio_H = function(Cu,Pu, Cd, Pd, u, d, r, S, type){
  if(type == 'call'){
    H = (Cu-Cd)/((u*S)-(d*S))
  } else{
    H =(Pu-Pd)/((u*S)-(d*S))
  }
  return(H)
}
call_replicate_porfolio_H(7.32, 0, 1.48, 1.56, 1.0488, 0.9747, 1.025, 100, 'call')
call_replicate_porfolio_H(7.32, 0, 1.48, 1.56, 1.0488, 0.9747, 1.025, 100, 'put')
```

The corresponding results in R confirm what we got above when we solved for H_0:

```
> call_replicate_porfolio_H(7.32, 0, 1.48, 1.56, 1.0488, 0.9747, 1.025, 100, 'call')
[1] 0.7881242
> call_replicate_porfolio_H(7.32, 0, 1.48, 1.56, 1.0488, 0.9747, 1.025, 100, 'put')
[1] -0.2105263
```

The R code for B_0,

```
risk_free_asset = function(Cu,Pu, Cd, Pd, u, d, r, S, type){
if(type == 'call'){
  B = (Cu*(d*S)-Cd*(u*S))/(r*((u*S)-(d*S)))
} else{
  B = (-(Pu*(d*S)-Pd*(u*S)))/(r*((u*S)-(d*S)))
}
return(B)
}
risk_free_asset(7.32, 0, 1.48, 1.56, 1.0488, 0.9747, 1.025, 100, 'call')
risk_free_asset(7.32, 0, 1.48, 1.56, 1.0488, 0.9747, 1.025, 100, 'put')
```

The corresponding results are,

```
> risk_free_asset(7.32, 0, 1.48, 1.56, 1.0488, 0.9747, 1.025, 100, 'call')
[1] 73.50094
> risk_free_asset(7.32, 0, 1.48, 1.56, 1.0488, 0.9747, 1.025, 100, 'put')
[1] 21.54146
```

The replicate portfolio V_u and V_d results in R,

```
Portfolio_replicate_call = function(H, B, u, d, r, S, type){
  if(type == 'up'){
    V = ((H*u*S)-(r*B))
  }else{
    V = ((H*d*S)-(r*B))
  }
  return(V)
}
Portfolio_replicate_call(0.7881242, 73.50094, 1.0488, 0.9747, 1.025, 100, 'up')
Portfolio_replicate_call(0.7881242, 73.50094, 1.0488, 0.9747, 1.025, 100, 'down')
```

The corresponding results are

```
> Portfolio_replicate_call(0.7881242, 73.50094, 1.0488, 0.9747, 1.025, 100, 'up')
[1] 7.320003
> Portfolio_replicate_call(0.7881242, 73.50094, 1.0488, 0.9747, 1.025, 100, 'down')
[1] 1.480002
```

The equilibrium price of the call in R is ,

```
attainable_call = function(H, B, S){
  Ca = ((H*S)-B)

return(Ca)
}
attainable_call(0.7881224, 73.50094, 100)
```

The corresponding results

```
> attainable_call(0.7881224, 73.50094, 100)
[1] 5.3113
```

6.2 Two-period case

For a two-period case (n=2), we have,

Upward movement for call option

If the call is not priced at \$ 7.32, an arbitrage portfolio could be constructed with a position in the call(long if the call is underpriced and short if it is overpriced) and an opposite position in the replicating portfolio with $H_u = 1$ and $B_u = \$97.56$.

$$H_u = \frac{C_{uu} - C_{ud}}{u^2 S_0 - udS_0} = \frac{10 - 2.23}{110 - 102.23} = 1$$

$$B_u = \frac{C_{uu}(udS_0) - C_{ud}(u^2 S_0)}{r(u^2 S_0 - udS_0)} = \frac{(10)(102.23) - (2.23)(110)}{1.025[110 - 102.23]} = \$97.56$$

$$C_u = H_u S_u - B_u = (1)(104.88) - 97.56 = \$7.32$$

Down movement for call option

When the share is priced at \$97.47, the call would have to be priced at $C_d = \$1.48$ or an arbitrage profit could be obtained by taking positions in the call and a replicating portfolio with $H_d = 0.3084$ and $B_d = \$28.586$

$$H_u = \frac{C_{ud} - C_{dd}}{udS_0 - d^2 S_0} = \frac{2.23 - 0}{102.23 - 95} = 0.3084$$

$$B_u = \frac{C_{ud}(d^2 S_0) - C_{dd}(udS_0)}{r(udS_0 - d^2 S_0)} = \frac{(2.23)(95) - (0)(102.23)}{1.025[102.23 - 95]} = \$28.57$$

$C_d = H_d S_d - B_d = (0.3084)(97.47) - 28.58 = \$1.48$

The R code for H_u,H_d,B_u,B_d in R is as follows

For H_u and H_d;

```
two_Step_H_call = function(Cuu, Cud, Cdd, u, d, r, S, type1){
  if(type1 == 'up call'){
    H_call = (Cuu-Cud)/(((u^2)*S)-((u*d)*S))
  }else{
    H_call =(Cud-Cdd)/((((u*d))*S)-((d^2)*S))
  }
return(H_call)
}
two_Step_H_call(9.998144, 2.226536, 0, 1.0488, 0.9747, 1.025, 100, 'up call')
two_Step_H_call(9.998144, 2.226536, 0, 1.0488, 0.9747, 1.025, 100, 'down call')
```

The corrosponding results from the above R function

```
> two_Step_H_call(9.998144, 2.226536, 0, 1.0488, 0.9747, 1.025, 100, 'up call')
[1] 1
> two_Step_H_call(9.998144, 2.226536, 0, 1.0488, 0.9747, 1.025, 100, 'down call')
[1] 0.3082766
```

In the case of the put option we have the following R function

```
two_Step_H_put = function(Puu, Pud, Pdd, u, d, r, S, type2){
  if(type2 == 'up put'){
    H_2_step = (Puu-Pud)/(((u^2)*S)-((u*d)*S))
  } else{
    H_2_step =(Pud-Pdd)/((((u*d))*S)-((d^2)*S))
  }
  return(H_2_step)
}

two_Step_H_put(0, 0, 4.9959910, 1.0488, 0.9747, 1.025, 100, 'up put')
two_Step_H_put(0, 0, 4.9959910, 1.0488, 0.9747, 1.025, 100, 'down put')
```

The corrosponding results from the above R function

```
> two_Step_H_put(0, 0, 4.9959910, 1.0488, 0.9747, 1.025, 100, 'up put')
[1] 0
> two_Step_H_put(0, 0, 4.9959910, 1.0488, 0.9747, 1.025, 100, 'down put')
[1] -0.6917234
```

The risk-free asset, B_u and B_d for both the call and the put option in R is evalauted as follows

```
risk_free_asset_call = function(Cuu,Cud, Cdd, u, d, r, S, type){
  if(type == 'up call'){
    B_call = (Cuu*((u*d)*S)-Cud*((u^2)*S))/(r*(((u^2)*S)-((u*d)*S)))
  } else{
    B_call = ((Cud*((d^2)*S)-Cdd*((u*d)*S)))/(r*(((u*d)*S)-((d^2)*S)))
  }
  return(B_call)
}

risk_free_asset_put = function(Puu,Pud, Pdd, u, d, r, S, type){
  if(type == 'up put'){
    I_put = -((Puu*((u*d)*S)-Pud*((u^2)*S))/(r*(((u^2)*S)-((u*d)*S))))
  } else{
    I_put = -((Pud*((d^2)*S)-Pdd*((u*d)*S)))/(r*(((u*d)*S)-((d^2)*S)))
  }
  return(I_put)
}

risk_free_asset_call(9.998144, 2.226536, 0, 1.0488, 0.9747, 1.025, 100, 'up call')
risk_free_asset_call(9.998144, 2.226536, 0, 1.0488, 0.9747, 1.025, 100, 'down call')
risk_free_asset_put(0, 0, 4.995, 1.0488, 0.9747, 1.025, 100, 'up put' )
risk_free_asset_put(0, 0, 4.995, 1.0488, 0.9747, 1.025, 100, 'down put' )
```

The corrosponding results in R is as follows

```
> risk_free_asset_call(9.998144, 2.226536, 0, 1.0488, 0.9747, 1.025, 100, 'up call')
[1] 97.56098
> risk_free_asset_call(9.998144, 2.226536, 0, 1.0488, 0.9747, 1.025, 100, 'down call')
[1] 28.57318
> risk_free_asset_put(0, 0, 4.995, 1.0488, 0.9747, 1.025, 100, 'up put' )
[1] 0
> risk_free_asset_put(0, 0, 4.995, 1.0488, 0.9747, 1.025, 100, 'down put' )
[1] 68.97411
```

The equilibrium price for both the call and the put in R is as follows

```
attainable_call = function(Hu, Hd, Bu, Bd, Su, Sd, type){
  if(type == 'up call'){
    ca = ((Hu*Su)-Bu)
  }else{
    ca =((Hd*Sd)-Bd)
  }

  return(ca)
}

attainable_put = function(Hu, Hd, Bu, Bd, Su, Sd, type){
  if(type == 'up put'){
    pa = ((Hu*(Su))+Bu)
  }else{
    pa =((Hd*(Sd))+Bd)
  }

  return(pa)
}
attainable_call(1, 0.3082766, 97.56098, 28.57318,104.8800,97.47000, 'up call')
attainable_call(1, 0.3082766, 97.56098, 28.57318,104.8800,97.47000, 'down call')
attainable_put(0, -0.6917234, 0, 68.97411, 104.88, 97.47, 'up put')
attainable_put(0, -0.6917234, 0, 68.97411, 104.88, 97.47, 'down put')
```

The corrosponding results in R are

```
> attainable_call(1, 0.3082766, 97.56098, 28.57318,104.8800,97.47000, 'up call')
[1] 7.31902
> attainable_call(1, 0.3082766, 97.56098, 28.57318,104.8800,97.47000, 'down call')
[1] 1.47454
> attainable_put(0, -0.6917234, 0, 68.97411, 104.88, 97.47, 'up put')
[1] 0
> attainable_put(0, -0.6917234, 0, 68.97411, 104.88, 97.47, 'down put')
```

7 Arbitrage Strategy

In the economy, if the market price of the call , C, is not equal to C_0, that is, $C > C_0 > H_0S_0 - B_0$ or $C < C_0 < H_0S_0 - B_0$. The arbitrage portfolio having a position in the call and an opposite position in the replicating portfolio can be constructed to take advantage of this mispricing opportunity. For example, if the call price in the above example , C_0, was \$6.31 instead of \$5.31, arbitrageurs could sell the costy call for \$6.31 and go long in the replicating portfolio (buy 0.7881 units of shares at \$100 and borrow \$ 73.501 dollors in a risk-free asset) earn a positive cash flow of \$1. At expiration, the cash flows from the short call position and the long replicating portfolio position would cancel each other out at either share price, and the initial profit of \$1 would be worth \$1.025. Thus, arbitrageurs would earn \$1.025 with no cash outflows. The more arbitrageurs try to sell calls at \$6.31, they will eventually push the price of the call down until it is equal to \$ 5.31. At that price, there won't be any arbitrage opportunity.

Likewise, if the call price is less than or below \$5.31, then the call is not expensive relative to the replicating portfolio $C < C_0 < H_0S_0 - B_0$. An arbitrage opportunity will exist by going long in the call and short in the replicating portfolio. For example, if $C = \$4.31$, then the arbitrageurs could buy the call ($-C = -\$4.31$) and take a short position in the replicating portfolio (sell 0.7881 of stock short at \$100 and invest \$73.501 dollars in risk-free security) to earn a positive cash flow or credit of \$1. At expiration, the cash flows from the long call position and the short replicating portfolio position would again cancel each other out at either stock price, and the initial profit of \$1 would be worth \$1.025. As the arbitrageurs try to buy the calls, the price will be pushed back to \$5.31 and the arbitrage opportunity will be gone.
In the binomial world, the arbitrage forces will consequently lead the price of the call to be equal to the value of the replicating portfolio. This would lead us to the assumption of no-arbitrage and complete market.

8 Conclusion

Think of the binomial option pricing models as a model in eaconomics used to study the demand and supply of goods in the economy. To reach equilibrium , the price of the good demanded must equal the price of the good supplied, and if that is not the case, then surplus or shortage of the goods will occur, causing the good's price to change until the equilibrium is reached. Similarly, the pricing of options using the binomial models is an attempt to study the up and down movement of share prices by trying to find a unique price at which the options tends to be.

In this paper we explored the binomial option pricing models from the build up of binomial trees to the valuation of option prices using the replication strategy. Our aim was to evaluate the binomial option pricing models in R, and to see if we can achieve similar results from the theoretical valuation which we done successfully.

9 references

Johnson, R. S. , 2009, Introduction to Derivatives, options and swaps, Oxford university Press, New York

Hull, J.C. , 2017, Options, Futures, and Other Derivatives, Global Edition, Pearson Education UK, UK

Swart, B., 2015, Financial Modelling course notes, University of South Africa (UNISA), South Africa

View publication stats